

KOLMARDEN-STROMSFORS, SwedenWMO: **025580**Lat: **58.689N**Lon: **16.307E**Elev: **509**StdP: **14.43**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.1	10.7	1.9	6.2	9.2	5.9	7.6	14.6	18.8	39.5	16.5	38.2	6.8	340	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	80.6	63.7	77.1	62.5	73.8	61.1	66.6	75.2	64.7	72.7	63.0	70.8	5.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	90.3	69.0	61.9	84.2	67.3	60.0	78.8	65.7	31.5	75.9	30.0	72.8	28.8	71.0	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.6	11.5	DB	3.4	86.5	5.6	4.3	-0.7	89.6	-4.0	92.1	-7.1	94.6	-11.2	97.7	(4)
(5)			WB	2.8	69.6	5.4	2.4	-1.1	71.4	-4.2	72.8	-7.3	74.2	-11.2	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	27.4	28.2	33.8	42.9	52.3	59.7	64.1	62.2	54.2	44.2	36.3	30.5	(6)	
	DBStd	14.56	8.00	7.74	7.42	6.61	6.41	5.59	5.47	4.75	5.31	6.20	6.47	8.21	(7)	
	HDD50	3324	699	610	503	231	47	2	0	0	22	195	410	605	(8)	
	HDD65	7495	1164	1030	967	664	396	178	85	113	325	643	860	1070	(9)	
	CDD50	1408	0	0	0	17	117	292	438	380	147	17	0	0	(10)	
	CDD65	105	0	0	0	0	2	17	58	27	1	0	0	0	(11)	
	CDH74	703	0	0	0	0	26	150	392	133	2	0	0	0	(12)	
	CDH80	125	0	0	0	0	1	25	80	19	0	0	0	0	(13)	
(14) Wind	WSAvg	6.1	6.7	6.5	6.5	5.9	5.6	5.8	5.7	5.5	5.9	6.2	6.5	6.5	(14)	
(15) Precipitation	PrecAvg	24.0	1.5	1.4	1.1	1.4	1.7	2.6	3.0	2.8	2.3	2.1	2.2	1.9	(15)	
	PrecMax	30.9	2.8	2.5	2.1	3.3	3.1	4.3	5.5	6.4	5.4	5.5	4.9	3.7	(16)	
	PrecMin	20.0	0.2	0.6	0.3	0.1	0.4	0.4	0.2	0.4	0.4	0.6	0.4	0.5	(17)	
	PrecStd	2.4	0.7	0.5	0.5	0.9	0.7	1.0	1.3	1.4	1.4	1.1	1.1	0.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	47.0	57.5	67.5	77.2	83.5	86.4	82.1	72.8	60.5	51.9	48.4	(19)	
		MCWB	43.0	42.3	46.4	53.0	58.0	62.3	66.2	65.7	60.9	55.1	49.3	46.1	(20)	
	2%	DB	41.9	43.0	51.7	62.5	72.3	78.2	82.1	77.5	68.9	57.1	49.2	44.4	(21)	
		MCWB	39.6	39.4	43.2	49.8	56.5	61.7	64.6	64.0	59.3	53.4	47.4	42.5	(22)	
	5%	DB	39.5	40.5	47.3	57.8	67.9	74.1	78.3	73.9	65.2	54.8	46.9	42.2	(23)	
		MCWB	37.8	38.2	40.9	47.6	54.8	59.6	63.6	62.5	57.5	51.9	45.5	40.8	(24)	
	10%	DB	37.4	38.3	43.7	53.8	63.8	70.6	74.8	70.9	62.2	52.6	44.8	40.2	(25)	
		MCWB	36.1	36.3	39.0	44.8	52.6	57.6	62.4	61.0	56.0	50.3	43.6	39.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	43.8	47.4	54.5	61.6	66.2	69.4	69.6	63.0	56.6	50.7	46.4	(27)	
		MCDB	45.0	45.9	55.8	65.0	70.8	75.0	79.3	77.2	69.1	59.0	51.3	48.3	(28)	
	2%	WB	40.2	40.5	44.1	51.0	58.8	63.2	67.2	66.2	60.4	54.3	47.8	43.0	(29)	
		MCDB	41.4	42.1	50.4	60.1	67.6	73.6	76.3	73.5	66.2	56.2	48.8	44.2	(30)	
	5%	WB	38.0	38.5	41.8	48.3	56.2	61.2	65.5	64.1	58.6	52.5	45.7	40.9	(31)	
		MCDB	39.2	40.4	46.3	56.7	64.7	71.4	74.5	70.3	63.6	54.2	46.6	41.8	(32)	
	10%	WB	36.1	36.3	39.6	45.7	53.9	59.3	63.7	62.6	56.9	50.8	43.8	39.0	(33)	
		MCDB	37.2	38.3	43.0	52.5	62.2	68.3	72.4	68.8	61.1	52.4	44.6	40.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	8.4	11.8	15.9	17.2	16.9	16.2	14.9	12.9	9.5	6.7	6.8	(35)	
		MCDBR	8.7	9.7	16.8	21.5	22.7	22.3	20.9	18.6	16.4	11.1	9.0	8.4	(36)	
	5% WB	MCWBR	8.3	7.4	10.4	11.9	11.4	10.1	8.0	8.0	8.6	7.8	8.3	8.2	(37)	
		MCWBR	8.5	9.1	15.6	20.2	20.2	20.0	18.4	16.2	15.0	10.3	8.2	8.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.278	0.315	0.333	0.370	0.366	0.355	0.381	0.373	0.341	0.332	0.302	0.263	(40)	
		taud	2.249	2.312	2.371	2.332	2.406	2.458	2.403	2.424	2.499	2.434	2.293	2.174	(41)	
	Ebn at Noon		175	225	257	265	275	279	268	262	254	218	163	137	(42)	
		Edh at Noon	14	22	27	34	33	32	33	30	24	19	14	11	(43)	
(44) All-Sky Solar Radiation	RadAvg		105	293	770	1261	1663	1840	1678	1334	892	400	136	72	(44)	
	RadStd		15	50	80	127	136	88	161	126	108	62	29	9	(45)	

Historical Trends

	DBAvg	Heating		Cooling
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